Augusto B. Corrêa

Junior Research Fellow, Christ Church, University of Oxford

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SUMMARY

Research scientist at the University of Oxford working on **how LLMs can acquire reasoning and planning capabilities through reinforcement learning, self-improvement, and interaction with formal systems.**

In ongoing research, I study how LLMs can self-improve their planning skills, and how agentic loops can be used to build better planning systems.

My research has been featured in the **Stanford AI Index Report 2026**. I have 20+ peer-reviewed publications, an award-winning Ph.D., and serve as **an invited expert on the Longitudinal Expert AI Panel (LEAP)**.

EXPERIENCE

Junior Research Fellow 11/2024 – present

Christ Church, University of Oxford, UK

- *Cross-Domain Transfer in LLM Reinforcement Learning (ongoing, 2026).* Trained LLMs with GRPO across different planning domains, comparing joint post-training against domain-specialist models. Joint training improves performance on most domains under matched training compute.
- *Iterative Deployment Improves Planning Skills in LLMs* (2025). Showed that iteratively fine-tuning LLMs on their own validated plans yields **1.5–5× more solved held-out tasks** after five generations.
- Built an end-to-end LLM training pipeline using PyTorch, LoRA/PEFT, and SFT, including data curation, fine-tuning, and evaluation on ∼600K validated agent traces.
- Showed that **LLMs can generate heuristic functions** that, when integrated into a real planner, **outperform decades of hand-crafted alternatives** (NeurIPS 2025, 2026).
- First-author article to appear in **Communications of the ACM**: frontier LLMs are becoming competitive with state-of-the-art planners.

Doctoral Researcher 01/2020 – 10/2024

University of Basel, Switzerland · Supervisor: Prof. Malte Helmert

- Designed search algorithms and heuristics for **Powerlifted**, a C++/Python planning system.
- Built large-scale evaluation pipelines for search systems.
- **Won the International Planning Competition 2023** with planners built on Powerlifted.
- Ph.D. awarded *summa cum laude*; thesis won the **ICAPS Outstanding Dissertation Award** (2025), the main AI planning venue.

Research Assistant 02/2018 – 12/2019

University of Basel, Switzerland

- Applied database query optimization to speed up planning systems (ICAPS 2020); published on heuristic search (ICAPS 2019) and learned heuristic functions with classical ML methods (IJCAI 2019)

EDUCATION

Ph.D. in Computer Science, University of Basel 2020 – 2024

Summa cum laude. **ICAPS Outstanding Dissertation Award** (2025).

M.Sc. in Computer Science, University of Basel 2018 – 2019

Major in Machine Intelligence.

B.Sc. in Computer Science, Federal University of Rio Grande do Sul 2013 – 2018

Best Bachelor's Thesis Award (Brazilian Computer Society).

SELECTED CODING-AGENT WORK

- **Verified agentic program synthesis (LeanPlan):** Co-developed an agentic system that generates planning heuristics and formally verifies their admissibility in Lean 4, guaranteeing optimal solutions when used with an optimal planner.
- **Automated Research for Planning:** Built autonomous coding-agent loops that implement research ideas and evaluate them on large-scale benchmarks, achieving **72% performance improvements** in existing planners.

SKILLS

LLM training	PyTorch, LoRA/SFT, GRPO, PEFT, TRL, distributed GPU training
LLM agents & reasoning	inference-time search, program synthesis, tool use, evaluation
Systems	Python, C++, Bash, Linux, Git, Slurm/HPC, large-scale evaluation pipelines
Open source	Powerlifted (C++/Python planner, creator and maintainer), pddl-skills (Python tools for LLM-based planning), llm-generated-heuristics (automated generation of heuristics using LLMs); contributor to Fast Downward and Downward Lab.

SELECTED PUBLICATIONS

1. **Corrêa, A.B.**; Gelberg, Y.; Melo, L.C.; Shumailov, I.; Pereira, A.G.; and Gal, Y. *Iterative Deployment Improves Planning Skills in LLMs.* arXiv:2512.24940. Under review at **ICLR 2027**.
   A model's own attempts, filtered by a pass/fail validator, suffice for large planning gains across model generations; training on all attempts works much worse.
2. **Corrêa, A.B.**; Pereira, A.G.; and Seipp, J. *Classical Planning with LLM-Generated Heuristics: Challenging the State of the Art with Python Code.* **NeurIPS 2025**.
   LLMs generate heuristic functions that outperform decades of hand-crafted alternatives inside a real planner.
3. **Corrêa, A.B.**; Pereira, A.G.; and Seipp, J. *Frontier Large Language Models Are Becoming Competitive with State-of-the-Art Planners.* **Communications of the ACM**, accepted; to appear.
   First paper to show that end-to-end planning with LLMs is competitive with established symbolic planners.
4. Pereira, A.G.; **Corrêa, A.B.**; and Seipp, J. *Property-Guided LLM Program Synthesis for Planning.* **NeurIPS 2026**, to appear.
   Self-improvement agentic loop where candidate programs are checked against formally defined properties, with counterexamples fed back to the LLM.
5. Pereira, A.G.; **Corrêa, A.B.**; Meneguzzi, F.; and Seipp, J. *LeanPlan: Optimal Planning with LLM-Generated Heuristics and Admissibility Proofs.* arXiv:2610.08246.
   Optimal planning with LLM-generated heuristics whose admissibility proofs are machine-checked in Lean 4.

SELECTED AWARDS AND SERVICE

- **Google's Gemini Academic Program Award** (2026): USD 10,000 for research on LLMs for planning.
- **ICAPS Outstanding Dissertation Award** (2025); **IPC Winner** (2023); **ICAPS Outstanding Program Committee Member Award** (2026).
- **ICAPS 2026 Best Paper Runner-Up**; **ICAPS 2023 Best Student Paper Runner-Up**; **ICAPS 2022 Best System Demonstration Award**.
- **Expert panel**: Transformative AI expert, Longitudinal Expert AI Panel (LEAP), Forecasting Research Institute (since 2026).
- **AI safety**: Finalist, Anthropic Fellows Program in AI safety (2026).
- **Workshop organizer**: LM4Plan (ICAPS 2025, ICAPS 2026, ICML 2026).
- **Programme committee**: NeurIPS, AAAI, IJCAI, ICAPS, KR, ECAI, SoCS.
- **Supervision**: 8 completed student theses (2 M.Sc., 6 B.Sc.).